

Basics 17: What Is a Plasma?

III - Elementary plasma physics

Ionization, quasineutrality, collective behavior, screening, coupling, and plasma regimes

Textbook draft, version 1.0. A plasma is not merely a hot gas containing ions. Its defining behavior is collective: many charged particles reorganize electric and magnetic fields over characteristic lengths and times. This chapter develops practical criteria for deciding when an ionized medium should be modeled as a plasma. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Define ionization fraction, quasineutrality, and collective behavior.
- Use Debye length and plasma parameter as plasma criteria.
- Distinguish weakly and strongly coupled plasmas.
- Explain why local charge separation can coexist with macroscopic quasineutrality.
- Classify common laboratory, space, and fusion plasmas.
- Identify when fluid, kinetic, or neutral-gas descriptions may be appropriate.

2 Prerequisites and reading strategy

- Basics 1, 10-13.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Ionized matter is necessary but not sufficient

A gas containing electrons and positive ions has ionization fraction

$$\chi_i = \frac{n_i}{n_i + n_n}, \quad (3.1)$$

where n_n is neutral density. A medium can exhibit plasma behavior even when $\chi_i \ll 1$ if charged particles respond collectively before neutral collisions erase that response. Conversely, a sparse collection of noninteracting charged particles is not automatically a plasma.

4 Collective behavior

Each charge interacts through long-range electromagnetic fields. In a plasma, the relevant response is not a sequence of isolated binary encounters but the coordinated rearrangement of many particles. Density disturbances can create plasma oscillations; currents can generate magnetic fields; pressure and magnetic tension can launch waves. Collective dynamics is meaningful when many particles occupy a shielding volume and system dimensions exceed the shielding length.

5 Quasineutrality and local charge separation

For singly charged ions,

$$n_e \simeq n_i, \quad (5.1)$$

so the net charge density

$$\rho_q = e(n_i - n_e) \quad (5.2)$$

is small compared with either species charge density. Quasineutrality is an asymptotic statement on scales much larger than the Debye length. It does not require exact equality point by point. Sheaths, double layers, waves, and small charge imbalances can generate important electric fields.

6 Debye length and system-size criterion

For electrons of temperature T_e and density n_e , the Debye length is

$$\lambda_D = \sqrt{\frac{\epsilon_0 k_B T_e}{n_e e^2}}. \quad (6.1)$$

A common plasma criterion is

$$L \gg \lambda_D, \quad (6.2)$$

so that an interior region can remain quasineutral while charge imbalances are screened over smaller scales. When L is comparable to λ_D , boundaries and electrostatic kinetics dominate.

7 Particles in a Debye sphere

The number of particles in a Debye sphere is

$$N_D = \frac{4\pi}{3} n_e \lambda_D^3. \quad (7.1)$$

The criterion $N_D \gg 1$ means that screening is a collective average over many particles and fluctuations are relatively small. It is closely related to weak coupling. Some texts define the plasma parameter using $n\lambda_D^3$ without the geometric factor; conventions should be stated.

8 Weak and strong coupling

Compare a characteristic Coulomb potential energy with thermal energy:

$$\Gamma = \frac{e^2}{4\pi\epsilon_0 a k_B T}, \quad a = \left(\frac{3}{4\pi n} \right)^{1/3}. \quad (8.1)$$

For $\Gamma \ll 1$, thermal motion dominates pair interaction energy and conventional weakly coupled plasma theory applies. For $\Gamma \gtrsim 1$, correlations become strong; dusty plasmas, dense matter, and some ultracold plasmas require different methods.

9 Magnetization and collisionality

A species is magnetized when it completes many gyrotations before collisions significantly scatter it and when its Larmor radius is small relative to system gradients:

$$|\Omega_s| \gg \nu_s, \quad \rho_s \ll L. \quad (9.1)$$

Electrons and ions can occupy different regimes simultaneously. Plasma classification therefore requires species, scale, and timescale labels rather than one global adjective.

10 Examples and modeling choices

Fusion plasmas are hot, magnetized, and often weakly collisional. The solar wind is dilute and collisionless but strongly collective. Fluorescent discharges can be partially ionized and collisional. Stellar interiors may be dense and collisional. A suitable model is selected by comparing λ_D , ρ_s , mean free paths, wave periods, and device dimensions.

11 Computational laboratory

Create a regime calculator that accepts density, temperature, magnetic field, system size, and collision frequency. Compute λ_D , N_D , Γ , gyrofrequencies, Larmor radii, and magnetization ratios. Apply it to a fusion core, glow discharge, solar wind, and ionosphere case.

12 Summary

- Plasma behavior is defined by collective electromagnetic response, not ionization alone.
- Quasineutrality is valid on scales large compared with the Debye length.
- Many particles per Debye sphere support a smooth collective description.
- The coupling parameter separates weakly and strongly correlated regimes.
- Model selection must be made for specific species, lengths, and times.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
χ_i	ionization fraction	Fraction of particles in the ionized component.
λ_D	Debye length	Electrostatic screening scale.
N_D	Debye-sphere population	Particles inside a shielding volume.
Γ	coupling parameter	Potential-to-thermal energy ratio.
a	Wigner-Seitz radius	Characteristic interparticle spacing.
ν_s	collision frequency	Rate of species- s scattering.

13 Exercises

1. Explain why an ionized gas with $L \ll \lambda_D$ may fail the usual plasma criterion.
2. Compute N_D for a specified electron density and temperature.
3. Show how λ_D changes when density is increased by a factor of four.
4. Distinguish quasineutrality from zero electric field.
5. Discuss why electrons may be magnetized while ions are not.

14 References

References

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